

COMPUTED TOMOGRAPHIC FINDINGS OF FUNGAL RHINITIS AND SINUSITIS IN CATS

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The computed tomographic (CT) findings of fungal rhinitis/sinusitis in cats were characterized. The CT images of 10 cats ranging in age from 7 to 13 years were examined. The mean age was 10.8 years and all were neutered males. Nasal aspergillosis was diagnosed in five cats, cryptococcosis in three cats, hyalohyphomycosis in one cat, and trichosporonosis in one cat. Bilateral disease was present in eight cats, seven had abnormal soft tissue attenuation in two-thirds of the nasal cavity, and six had turbinate lysis. Seven cats had also lysis of the hard palate, nasal septum, or frontal bone. One cat had lysis of the cribriform plate. Five of the nine cats whose lymph nodes were imaged had lymph node enlargement. There was contrast medium enhancement in the nasal cavity in all cats, with either a primarily peripheral rim or heterogeneous pattern. There appears to be an overlap of clinical signs, age, and CT features of cats with nasal neoplasia and those with fungal rhinitis/sinusitis. *Veterinary Radiology & Ultrasound*, Vol. 50, No. 1, 2009, pp 65–68.

Key words: cat, computed tomography, feline, fungal, nasal, rhinitis, sinusitis.

Introduction

THERE ARE VARIOUS causes of feline nasal disease that lead to nasal discharge.^{1–4} Imaging is important in attempting to establish the cause of nasal discharge.^{1–6} However, in many patients establishing a cause is not possible. For example, a definitive diagnosis was found in only 27 (36%) of 75 cats with chronic nasal discharge.⁴

Cryptococcus neoformans is a common cause of feline nasal mycotic disease.^{7,8} Although not as frequent, *Aspergillus* spp. has also been implicated in feline nasal disease.^{7,8} Other fungal diseases affecting the nasal cavity in cats include hyalohyphomycosis and phaeohyphomycosis.⁸

Certain computed tomographic (CT) characteristics have been associated with specific nasal disease.^{5,9–13} Although the CT findings of nasal aspergillosis in the dog have been characterized, less is known about the CT findings of feline nasal mycotic disease. The purpose of this retrospective study was to describe the CT features of feline nasal mycoses.

Materials and Methods

CT examinations of 10 cats with nasal mycotic disease were performed at one of our institutions between Decem-

ber 2005 and March 2008. Patients had nasal discharge and/or ocular discharge ranging from 2 days to 1 year before imaging. Fungal rhinitis was confirmed using histopathologic, serologic, and culture results.^{2,3} Rhinoscopy was performed when available.

The mean age was 10.8 years with a range from 7 to 13 years. There were five domestic short hair, three Persian, one domestic long hair, and one Siamese. All were neutered males. Nasal aspergillosis was diagnosed in five cats, cryptococcosis in three cats, hyalohyphomycosis in one cat, and trichosporonosis in one cat.

CT images were obtained using either a helical CT scanner*,†,‡ (slice thickness 1–3 mm, kV 120, mA 100–180, pitch 1.5) or a conventional CT scanner§ (slice thickness 3 mm, kV 120, mA 80). Transverse images of the skull were obtained perpendicular to the hard palate. Nonionic iodinated contrast medium was given as a hand-injected intravenous bolus¶,|| using a dosage of 480–600 mg iodine/kg.

Digital images were jointly evaluated by two reviewers (K.K. and J.R.) in bone and soft tissue windows. The bone window width (WW) was 2500 and bone window level (WL) ranged from 480 to 500. The soft tissue WW was 350 and WL ranged from 40 to 50. These values were adjusted as needed to better evaluate the images.

CT criteria included: laterality of soft tissue attenuation within the nasal cavity (uni- or bilateral); amount of abnormal soft tissue attenuation within the nasal cavity

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†HiSpeed CT/i, GE Medical Systems, Milwaukee, WI.
‡Highlight CT, GE Medical Systems, Milwaukee, WI.
§Sytek 3000 CT, GE Medical Systems, Milwaukee, WI.
¶Optimark, GE Healthcare, Princeton, NJ.
||Isovue, Bracco Diagnostics, Princeton, NJ.

(none, minimal, moderate, large); extension of lesions (rostral, middle, caudal nasal cavity); frontal sinus involvement (none, present); turbinate lysis (none, mild, moderate, severe); osseous changes (location, destruction, sclerosis); tympanic bulla involvement (none, present); nasopharyngeal involvement (none, present); predominant contrast medium enhancement pattern (none, homogeneous, heterogeneous, peripheral); and lymph node involvement (none, enlarged, contrast enhancement pattern).^{3,10-12}

Pertaining to the nasal passages, the term none was used when there was no evidence of disease, mild/minimal was used when disease affected less than one-third of the nasal cavity, and moderate when approximately two-thirds of the nasal cavity was involved.¹⁰ The term severe/large referred to disease affecting more than two-thirds of the cavity. The term rostral was used to describe disease confined rostral to the roots of the maxillary canine teeth and middle was used when disease extended to the level of the fourth maxillary premolar teeth. The caudal nasal cavity was defined as the portion of the nasal cavity caudal to the fourth maxillary premolar teeth. Lymph node enlargement and laterality were subjectively assessed.

Results

Disease of the nasal cavity was bilateral in 8/10 patients and only the right side was affected in 2/10 cats. In these two cats, the amount of abnormal soft tissue in the nasal cavity was minimal. There was a focal area of soft tissue attenuation ventrally along the right frontal bone adjacent to the orbit in one cat and mixed soft tissue and air attenuation extending from the mid to caudal nasal cavity in the other cat (Fig. 1).

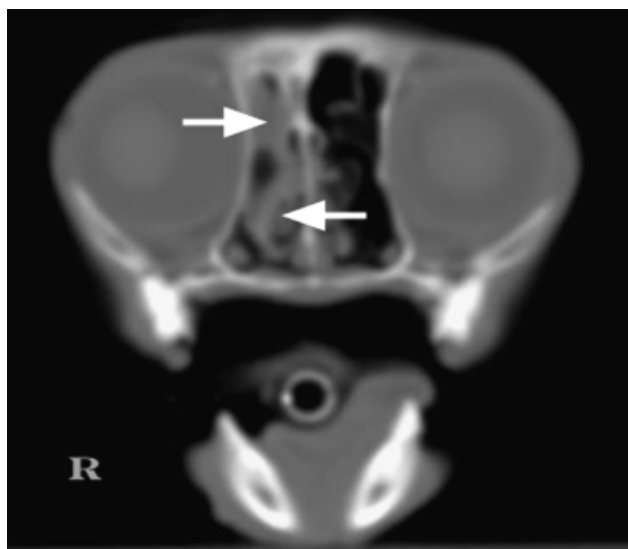


FIG. 1. Transverse computed tomographic (CT) image (window width = 2500, window level = 480) of the nasal cavity at the level of the orbits in an 11-year-old Persian cat with aspergillosis. There is increased soft tissue attenuation in the right nasal cavity (arrows).

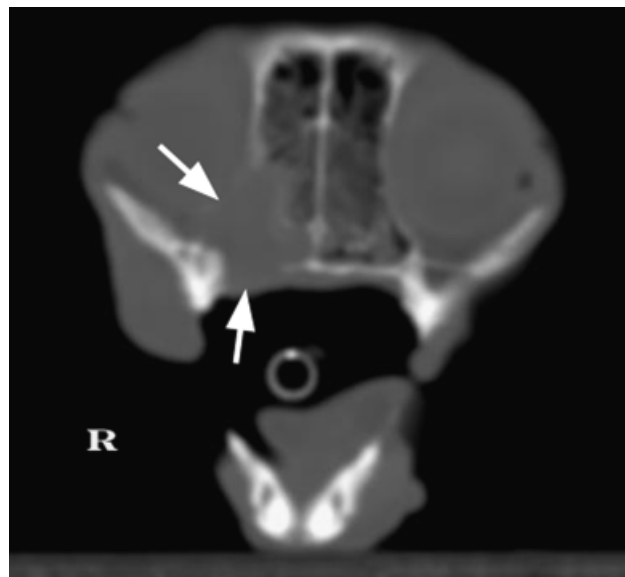


FIG. 2. Transverse computed tomographic (CT) image (window width = 2500, window level = 480) of the nasal cavity at the level of the orbits in an 8-year-old domestic shorthair cat with aspergillosis. There is a focal soft tissue attenuating mass effect (arrows) in the right ventral nasal cavity with extension into the right orbit. Lysis of the adjacent hard palate and right frontal bone is also present.

In the eight cats with bilateral disease, seven cats had moderate amounts of abnormal soft tissue attenuation in the nasal cavity. The right side was more severely affected compared with the left side in 4/7 cats and there was symmetric involvement in the remaining three cats. In 1/8 cats, the amount of abnormal soft tissue was large and the left side was more severely affected than the right.

There were no cats with lesions confined to the rostral or caudal portions of the nasal cavity. In 1/10 cats, the nasal cavity was affected from the rostral to the middle portions. In 5/10 patients, nasal disease extended from the rostral to caudal nasal cavity and from the middle to caudal nasal cavity in 2/10 cats. Disease was confined to the middle region of the nasal cavity in 2/10 cats.

Turbinate lysis was present in 6/10 cats. Mild turbinate lysis was seen in 2/6 cats, moderate in 3/6 cats, and severe in 1/6. In 1/10 cats, the nasal turbinates were obscured by soft tissue and could not be adequately assessed. Lytic lesions of the palatine bone, nasal septum, frontal bone, and/or cribriform plate were present in 8/10 patients, one of which had a focal soft tissue mass (Fig. 2). Lysis of the hard palate was present in 3/8 cats and lysis of the nasal septum was evident in 3/8 cats. Frontal bone lysis was seen in 4/8 cats. There was lysis of the cribriform plate in one cat despite the absence of an adjacent soft tissue mass. Soft tissue attenuation with a fluid line was present in the right tympanic bulla of 1/10 cats. Soft tissue attenuation was also present in the rostral aspect of the nasopharynx in this cat, primarily left sided. The tympanic bullae were unremarkable in all other cats.



FIG. 3. Transverse computed tomographic (CT) image (window width = 350, window level = 40) of the caudal nasopharynx in a 13-year-old domestic long hair cat with cryptococcosis. There is a large, well-margined, nasopharyngeal soft tissue attenuating mass (arrows) with a peripheral rim of contrast medium enhancement and central hypoattenuation. This mass causes ventral deviation of the soft palate.

The frontal sinuses were affected in 5/10 cats, bilaterally in 3/5 cats. Only the right frontal sinus was affected in 2/5 cats, which were also the cats with unilateral, right-sided disease of the nasal cavity. There were varying amounts of soft tissue attenuation within the frontal sinus in these cats. Enhancement was seen in the frontal sinus of 3/5 cats following administration of contrast medium. A peripheral contrast medium enhancement pattern was seen in 2/3 cats and a heterogeneous pattern in 1/3 patients.

The nasopharynx was affected in 5/10 cats. This consisted of mixed soft tissue/air attenuation in 4/5 cats. In one cat, a distinct nasopharyngeal mass effect was present,

causing ventral deviation of the soft palate (Fig. 3); this was proven to be a fungal granuloma due to *Cryptococcus*. Contrast medium enhancement was present in 4/5 cats with a homogeneous pattern in 1/4 cats and heterogeneous pattern in 1/4 cats. A peripheral rim of enhancement with central hypoattenuation was present in 2/4 cats, including the patient with the nasopharyngeal mass.

Contrast medium enhancement was present in the nasal cavity of all 10 cats. Abnormally increased peripheral enhancement of the mucosal lining was present in 4/10 cats. Heterogeneous enhancement was evident in 4/10 patients and homogeneous contrast medium enhancement within the abnormal soft tissue was present in 2/10 cats.

The mandibular and retropharyngeal lymph nodes were imaged in 9/10 cats. Enlarged lymph nodes were present in 5/9 cats (Fig. 4). The right mandibular and retropharyngeal nodes were enlarged and heterogeneously contrast enhancing with central hypoattenuation in 1/5 cats and homogeneously enhancing in 1/5 cats. In 3/5 cats, only the retropharyngeal nodes were enlarged. Heterogeneous contrast medium enhancement with central hypoattenuation was present in 2/3 cats and homogenous enhancement of the retropharyngeal node was seen in 1/3 cats. In cats with unilateral nasal disease, there was no lymph node enlargement.

Discussion

Imaging findings associated with feline neoplasia have been compared with those in other sinonasal disease. Some CT features suggestive of a neoplastic process compared with a rhinitis include osteolysis of paranasal bones, moderate to severe turbinate destruction, lysis of the nasal septum, a homogenous space occupying mass, and extension of disease into the orbit or facial soft tissues.³ No cat with rhinitis had lysis of the nasal septum.³ This is in contrast to our study, where lysis of the nasal septum was seen in three

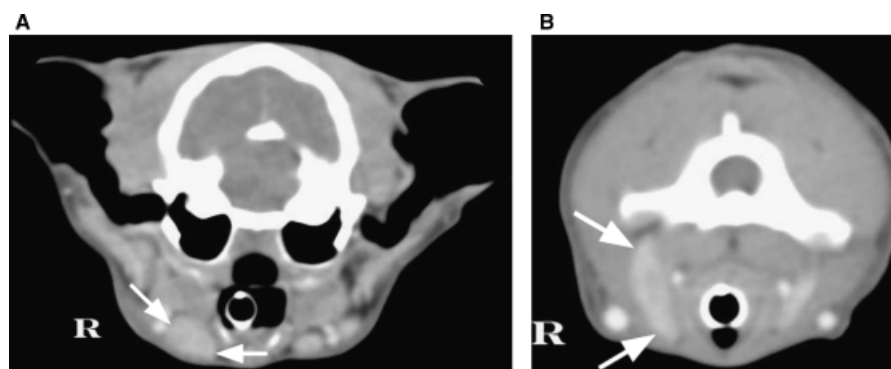


FIG. 4. Transverse computed tomographic (CT) images (window width = 350, window level = 40) of lymphadenopathy with heterogeneous contrast medium enhancement in a cat with aspergillosis. (A) Enlargement of the right mandibular lymph node (arrows). A biopsy of the node revealed *Aspergillus* hyphae. (B) Enlargement of the right retropharyngeal lymph node (arrows) with central hypoattenuation.

cats. A nasopharyngeal mass has also been associated with neoplasia.⁵

The mean age of cats with a nasal tumor has been reported as 11 years³ or 11.3 years,⁵ compared with the mean age of cats with inflammatory disease of 8.1 years³ or 9.0 years.⁵ In our study, the mean age for cats with fungal rhinitis was 10.7 years, which appears similar to patients with a nasal tumor. In contrast, maxillary bone lysis and bilateral lysis of the orbital lamina were highly correlated with neoplasia and all cats with neoplasia had bilateral disease.⁵ None of our patients has maxillary or bilateral orbital lamina lysis and 2/10 cats had unilateral disease.

In feline rhinitis, the frontal bone was the most commonly affected bone.³ This is consistent with our findings in which frontal bone lysis was present in half of the cats with lytic lesions. However, a difference is that bone destruction was only present in 7/25 (28%) of cats with non-fungal rhinitis compared with 8/10 in this study.³ Interestingly, all 10 cats with confirmed nasal/sinonasal mycotic disease were male, whereas in a previous study of cats with nasal neoplasia and rhinitis, male and females were equally represented.³

In dogs, nasal aspergillosis has a relatively specific appearance including cavitated turbinate lysis, a rim of soft tissue along the bones of the frontal sinus, maxillary recess and nasal passages, soft tissue accumulation, and reactive bony changes.^{10–13} These were not seen in cats with nasal aspergillosis in this study. Although it is inaccurate to compare the CT findings of feline nasal aspergillosis with cryptococcosis due to the limited number of cats, it is in-

teresting that the only cat with a distinct nasopharyngeal mass had cryptococcosis. Although sinonasal aspergillosis is rare in cats, 5/10 cats in this study had confirmed aspergillosis compared with 3/10 with cryptococcosis.⁸ However, many cats test positive for cryptococcosis via serum titer tests, eliminating the need for CT imaging. It is interesting that the cat with the largest amount of abnormal soft tissue in the nasal cavity and the greatest degree of turbinate destruction had trichosporonosis. This cat also had soft tissue attenuation in the right tympanic bulla.

Because of the low number of cats in this study, it is difficult to make concrete conclusions for specific CT findings of feline fungal rhinitis/sinusitis, but it does appear that some of the features overlap with those seen in sinonasal neoplasia, including age at presentation, soft tissue mass effect, and osteolysis. There are also similarities with nonfungal rhinitis including the predominance of frontal bone lysis; however, the number of cats with bone lysis seems higher in cats with mycotic disease. The predominance of male cats, the fact that the majority of the cats were diagnosed with feline nasal aspergillosis, and the number of Persian cats in our study warrants further investigation. Subsequent studies with a larger number of cats should be performed to better characterize differences between fungal rhinitis and other sinonasal disease and potentially be subcategorized for specific mycotic infections.

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